Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18SE7DE53]

SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER-2021

SOFTWARE PROJECT MANAGEMENT

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1. a. Paul Duggan is the manager of a software development section. On Tuesday at 10.00 AM and his fellow section heads have a meeting with their group manager about the staffing requirements for the coming year. Paul has already drafted document 'bidding' for staff. This is based on the work planned for his section for the next year. The document is discussed at the meeting. At 2.00 p.m. Paul has a meeting with his senior staff about an important project his section is undertaking. One of the programming staff has just had a road accident and will be in hospital for some time. It is decided that the project can be kept on schedule by transferring another team member from less urgent work to this project. A temporary replacement is to be brought in to do the less urgent work but this may take a week or so to arrange. Paul has to phone both the human resources manager about getting a replacement and the user for whom the less urgent work is being done, explaining why it is likely to be delayed? List all the eight management responsibilities Paul was responding to at different points during his day and identify whichever matching to the scenario. **08**

b. Calculate Return on Investment for the following scenario where company has average annual profit of 25000 and total investment is 630000. **06**

2. a. Sketch and explain the incremental delivery and advantages of this approach **10**

b. Sketch the product breakdown structure for a system development task. **04**

3. a. Calculate the risk exposure, Say a project depended on a data centre vulnerable to fire. It might be estimated that if a fire occurred a new computer configuration could be established for $500,000. It might also be estimated that where the computer is located there is a 1 in 2 chance of a fire actually happening, that is a probability of 0.5. **07**

b. Solve the Euclidean distance and explain Estimating by analogy between two projects, the new project is known to require 7 inputs and 15 outputs. One of the past cases, project A, has 8 inputs and 17 outputs. **07**

OR

4.	a.	Compare Organic mode, Embedded mode and Semi detached mode	06
	b.	Classify the Software effort estimation techniques and list the Estimation Difficulties.	08
5.	a.	Identify the major seven resources that are required for completing a full e-commerce web application from the scratch.	10
	b.	Calculate the schedule variance where PV is 40,000 and EV is 35,000.	04
		OR	
6.	a.	Compare three different earned value analysis techniques such as 0/100 technique, The 50/50 technique, The 75/25 technique;	06
	b.	Sketch the Project reporting structures and list the responsibility of project steering committee.	08
7.	a.	Summarize the The Oldham–Hackman job and methods of improving motivation.	10
	b.	Classify the different leadership styles that can be involved in carrying out a project.	04
		OR	
8.	a.	Classify the different types of people that Belbin suggested for balancing the project in an organization.	10
	b.	Tabulate time and place constraints by appropriate communication methods.	04